



Concurrent Strength and Endurance Training in Elderly

Biological aging is associated with declines in the neuromuscular and cardiovascular systems, resulting in an impaired capacity to perform daily activities [1–5]. In addition, age-related muscle power decrease is also an important predictor of functional limitations in healthy elderly [6–12].

Strength and endurance training promote specific neuromuscular and cardiovascular adaptations. The adaptations induced by strength training include muscle hypertrophy [13, 14] and an increase in both the motor unit recruitment capacity and motor unit firing rate [15–17]. These neuromuscular adaptations result in improved muscle strength and power development [18]. In contrast, endurance training induces central and peripheral adaptations that improve the cardiovascular function and the capacity of skeletal muscles to generate energy via the oxidative metabolism [19, 20]. Thus, a combination of strength and endurance training (i.e., concurrent training) in elderly populations is the most effective way to enhance both neuromuscular and cardiorespiratory functions, and consequently to preserve functional capacity [21–25]. In addition, another benefit of prescribing both strength and endurance training concurrently in elderly is that, in order to perform both modalities, the total time spent undergoing physical activity is increased, which is also beneficial to prevent and control cardiometabolic diseases [26]. However, as has been described in previous chapters of the book, the simultaneous development of both neuromuscular and cardiovascular adaptations might be challenging; and, in order to optimize the concurrent training prescription, it seems relevant to identify

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the most effective combination of training variables (i.e., intensity, volume, weekly frequency, exercise order) to promote both neuromuscular and cardiovascular adaptations in the elderly.

Table 18.1 shows a summary of the methods and results from studies which compared neuromuscular, cardiorespiratory, and functional adaptations to strength, endurance, and concurrent training, as well as different concurrent training regimes in elderly populations (i.e., mean age of above 60 years).

Effects of Concurrent Training on Muscle Strength, Rate of Force Development, and Power Output

In aging populations, most of the studies reported that concurrent training induced similar strength adaptations using two sessions per week of each modality (i.e., strength and endurance) when compared with strength training alone [34–39]. However, we have previously demonstrated that three times a week of concurrent training can result in an interference effect in elderly men [25]. In addition, the time-course of strength development during a concurrent training periodization may be influenced by the weekly training frequency [40, 41]. Furthermore, we have also shown that the intra-session exercise sequence may influence the magnitude of strength adaptations in the elderly, and performing strength training prior to endurance exercise may optimize the neuromuscular adaptations in this population [27, 28].

In previous study of our laboratory, Izquierdo et al. [22] investigated the effects of 16 weeks of strength, endurance, and concurrent training among elderly men. In this study, the strength and endurance training groups performed specific training twice a week, and the concurrent training group performed strength exercises on one day and cycle ergometer training on the other day. This study demonstrated that after 16 weeks of training, similar lower-body strength gains were observed in the strength and concurrent training groups, which suggests that a minimum weekly frequency of concurrent training may promote an optimal stimulus to strength gains in previously untrained elderly subjects [22].

When similar volumes of strength training only and concurrent training were performed on separate days (i.e., four training sessions per week), Karavirta et al. [39] observed similar strength gains (i.e., 21 and 22% in dynamic leg press) and similar improvements in concentric leg extension power (~16%) after 21 weeks of training twice a week in 40–67-year-old men. Using a similar training volume, intensity, and weekly frequency, other studies have shown comparable strength and power gains induced by strength and concurrent training in older men [35–37, 39], and older women [38].

However, increasing the weekly training frequency of concurrent training from two to three sessions per week may induce an interference effect in elderly men. This was shown by our study in elderly men (65 ± 5 years), where 12 weeks of training performed three times a week led to greater dynamic and isometric strength in the leg extensor muscles in the group that performed only strength training (dynamic strength: 67%; isometric strength: 14%) when compared with a combined strength

Table 18.1 Neuromuscular, functional, and cardiorespiratory adaptations to endurance-, strength-, and concurrent training in the elderly

Authors	Period (weeks)	Weekly frequency	Volume and intensity	Main findings
Wood et al. [21]	12	ST: 3 x/wk; ET: 2 x/wk; CT: ST + ET	ST: 2 sets, 12–15 rep (75% 5 RM) of 8–12 RM ET: 60–70% of HR _{max} estimated, 20–45 min cycling CT: 1 set of 8–12 RM + 30 min cycling	ST: ↑ 5 RM (44%) CT: ↑ 5 RM (38%) No differences between CT and ST in 5 RM gains. All groups: ↑ functional fitness, but greater ↑ in CT
Izquierdo et al. [22]	16	ST: 2 x/wk; ET: 2 x/wk; CT: 1 x/wk ST + 1 x/wk ET	ST: 3–5 sets, 6–15 rep (50–80% 1 RM) slow + fast contractions (20% of total volume, 30–50% of 1 RM) ET: 30–40 min cycling, at loads (W) of 2, 3, and 4 mmol L ⁻¹ of lactate	ST: ↑ 1 RM (41%); ↑ muscle power (37%). ↑ CSA QF (11%). CT: ↑ 1 RM (38%); ↑ muscle power (38%). ↑ CSA QF (11%). No differences among CT and ST in strength and CSA gains. ↑ W _{max} (16%) in ST, CT, and ET groups
Cadore et al. [25]	12	ST: 3 x/wk; ET: 3 x/wk; CT: ST + ET, ET prior to ST at same session	ST: 18–20 RM progressing to 6–8 RM ET: 20–30 min cycling, 80–100% of HR at VT ₂	ST: ↑ 1 RM (67%); ↑ PT (14%); ↑ EMG QF (30%). CT: ↑ 1 RM (41%); ST had greater increases in the 1 RM, PT, and EMG than CT. ↑ VO _{2peak} (20%); ET: ↑ VO _{2peak} (20%); ↑ 1 RM (22%)

(continued)

Table 18.1 (continued)

Authors	Period (weeks)	Weekly frequency	Volume and intensity	Main findings
Cadore et al. [24]	12	ST: 3 x/wk; ET: 3 x/wk; CT: ST + ET, ET prior to ST at same session	ST: 18–20 RM progressing to 6–8 RM ET: 20–30 min cycling, 80–100% of HR at VT ₂	ST : ↑ dynamic NEURO ECO at 100 W of VL (14%); CT e ET : ↑ W _{max} (19%); ↑ dynamic NEURO ECO at 100 W of VL (10%) and 50, 75, and 100 W of RF (14–35%) SE : ↑ force per unit of muscle mass (27%); ↑ MT QF (9%); ES : ↑ force per unit of muscle mass (15%); ↑ MT QF (9%); Greater changes in the SE than in the force per unit of muscle mass (27 vs. 15%)
Cadore et al. [27]	12	ST: 3 x/wk; ET: 3 x/wk; CT: ST + ET at same day	ST: 18–20 RM progressing to 6–8 RM ET: 20–30 min cycling, 80–100% of HR at VT ₂ CT performed with ET performed prior to (ES group) vs. after ST (SE group)	SE : ↑ 1 RM (35%); ↑ PT (8%)*; ↑ MT of VL, VM, RF, and VI (4–16%); ↑ QF EMG (~20%); ↑ NEURO ECO of RF in SE (22%). ES : ↑ 1 RM (21%); ↑ PT (5.7%); ↑ MT of VL, VM, RF and VI (4–16%); ↑ QF EMG (~20%). Greater changes in the SE than in the 1 RM (35 vs. 21%)
Cadore et al. [28]	12	ST: 3 x/wk; ET: 3 x/wk; CT: ST + ET at same day	ST: 18–20 RM progressing to 6–8 RM ET: 20–30 min cycling, 80–100% of HR at VT ₂ CT performed with ET performed prior to (ES group) vs. after ST (SE group)	↑ upper and lower-body 1 RM (35%) in both groups; ↑ QF MT in both groups (8%); ↑ VO _{2peak} in both groups; ↑ W _{max} only in CT3
Ferrari et al. [30]	10 after previous 20 weeks of training	2 x/wk (CT2) vs. 3 x/wk (CT3) at same day	ST: 10–12 RM progressing to 6–8 RM ET: 30 min cycling, 85 progressing to 95% of HR at VT ₂ CT performed with ST performed prior to ET	

Table 18.1 (continued)

Wilhelm et al. [31]	12	ST: 2 x/wk; ET: 2 x/wk; CT: ST + ET at same day	ST: 15–18 RM progressing to 8–10 RM ET: 20–40 min cycling, 85–95% of HR at VT ₂ CT performed with ET performed prior to (ES group) vs. after ST (SE group)	↑ 1 RM; ↑ peak power; ↑ MT QF; ↑ RFEI; ↑ 30SS*; ↑ TUG performance; ↑ QF EMG; No differences between SE and ES
Ferrari et al. [32]	10 after previous 20 weeks of training	2 x/wk (CT2) vs. 3 x/wk (CT3) at same day	ST: 10–12 RM progressing to 6–8 RM ET: 30 min cycling, 85 progressing to 95% of HR at VT ₂ CT performed with ST performed prior to ET	↑ isokinetic PT at 60 and 180° s ⁻¹ ; ↑ force per unit of muscle mass; ↑ CMJ height; No differences between groups
Da Silva et al. [33]	12	2 x/wk	ST: 65% progressing to 75% of 1 RM. RFG performing repetitions to concentric failure; NFG performing 50% of possible maximal repetitions; ENFG performing 50% of possible maximal repetitions but doubling the number of sets ET: 20–30 min at 60–65% of maximal HR, progressing to 70–75%	↑ 1 RM similarly among groups; ↑ force per unit of muscle mass; similarly among groups; ↑ RTD similarly among groups; ↑ SJ and CMJ performance similarly among groups; Greater ↑ MT QF in ENFG and RFG compared to NFG

↑ increases; *min* minutes; *NEURO ECO* neuromuscular economy; *1 RM* lower-body one maximum repetition; *PT* isometric peak torque; *CSA* cross-sectional area; *MT* muscle thickness; *QF* quadriceps femoris; *VL vastus lateralis*; *VM vastus medialis*; *RF rectus femoris*; *VI vastus intermedius*; *EI* echo intensity; *30SS* 30 second sit-to-stand test; *TUG* timed-up-and-go-test; *SJ* squat jump; *CMJ* counter movement jump; *x/wk* training sessions per week; *ST* strength training; *ET* endurance training; *CT* concurrent training; *ES* endurance prior to strength training; *SE* strength prior to endurance training; *RFG* repetitions to failure group; *NFG* repetitions not to failure group; *ENFG* repetitions not to failure group with equalized volume; *EMG* electromyographic signal; *VT₁* and *VT₂* first and second ventilatory threshold, respectively; *VO_{2peak}* peak of oxygen uptake; *W_{max}* maximal power at cycle ergometer; *HR* heart rate. Significant difference: All mentioned differences are statistically significant

and cardiovascular group (dynamic strength: 41%; isometric strength: 2%), whereas similar upper-body strength gains were evidenced in the strength and concurrent training groups (30–33%) [25]. Based on these results, we suggested that the interference effect of endurance training on strength adaptations occurs only in the specific muscle groups that perform both strength and endurance exercises (i.e., lower-limbs). Although an interference effect was observed in the concurrent training group, this group exhibited a similar magnitude of strength gains in relation to the results of the abovementioned studies (i.e., approximately 20–30%) [34, 35, 37–39], and the same strength adaptations occurred in a shorter period of time (12 vs. 21 weeks). These different time courses in strength development could be explained by the different weekly frequencies of training performed. In our study [25], subjects performed three training sessions per week, while in other previous studies only two training sessions per week were performed (i.e., ~30% lower volume) [34, 35, 37–39].

In contrast, we have previously shown that a weekly frequency of three times a week did not promote greater maximal strength [30] and jump performance [32] gains in well-trained healthy elderly subjects when compared with twice per week (20–22%, respectively). In this study, we suggested that in previously concurrently trained elderly subjects, twice per week may be an optimal weekly frequency to enhance muscle strength and jump performance. However, in another study on elderly women, Fisher et al. [42] have shown that 16 weeks of 1, 2, and 3 sessions of endurance and strength training lead to similar improvements in muscle strength, cardiovascular fitness, and functional tasks performance. Therefore, it seems that the recommended weekly frequency of concurrent training may differ for elderly men and women. However, this issue needs to be further investigated.

Another factor that may influence the magnitude of strength adaptations in the elderly is the intra-session exercise sequence. Greater maximal dynamic strength gains (35 vs. 21%) and greater force per unit of muscle mass (27 vs. 15%) were observed in a concurrent group that performed strength training prior to endurance exercise, when compared with the inverse order [27, 28], after 12 weeks of concurrent training using a similar training periodization model, which had induced an interference effect in our previous study [25]. It may, thus, be suggested that the lower strength gains obtained after the endurance-strength exercise sequence could be related to lower workloads used during the training in this group, which probably was related to residual fatigue induced by high-intensity endurance training on the cycle ergometer, especially during the last 4 weeks of training [28]. Regarding explosive force, no differences in the maximal rate of force development (RFD) and that of 100 ms were observed when comparing the two exercise sequences, with both achieving similar improvements after 12 weeks of training. Conversely, Wilhelm et al. [31] observed similar strength and power output gains, irrespective of the exercise order. However, some differences between studies by Cadore et al. [28, 29] and Wilhelm et al. [31] may explain the discrepant results: in our studies [27, 28], the weekly frequency was three times per week, endurance training intensity achieved 100% of second ventilatory threshold (VT_2), and strength training was performed using 6–8 RM in the last 2 weeks. On the other hand, in the study by Wilhelm et al.

[31], the weekly frequency was twice per week, endurance training intensity achieved 95% of VT_2 , and strength training was performed using 8–10 RM in the last 2 weeks, which probably means a lower strength training intensity in the final phase of training. Along with differences in the training intensity in both modalities (i.e., strength and endurance), the different weekly frequencies related on average to approximately a 50% higher training volume in the knee and hip extensors when training was performed three times per week [27, 28], which may have caused these different results. Since in young men and women no difference was observed between the two exercise orders following 24 weeks of training [43], it might well be suggested that elderly individuals may be more sensitive to residual fatigue during the subsequent strength training. Importantly, however, recently, we have shown that concurrent training performed with repetitions to concentric failure did not provide additional gains in the knee extensors maximal strength, explosive strength, in healthy elderly individuals. In addition, even when performing 50% of the possible maximal repetitions with no volume compensation, elderly men optimized maximal and explosive strength gains within 12 weeks of training [33]. From a practical standpoint, the performance of submaximal sets promoted the same magnitude of strength enhancements, showing that this is an efficient alternative for improving neuromuscular function in elderly men [33].

Effects of Concurrent Training on Muscle Hypertrophy

Although several studies have investigated the effects of strength training on muscle mass in older subjects, a lower number of studies have explored the effects of concurrent training on muscle hypertrophy in this population. In the study by Izquierdo et al. [22], no differences were observed between the strength (twice a week) and concurrent training groups (one session per week of strength and one session per week of cycle endurance training) in the magnitude of hypertrophy after 16 weeks of training (increases of ~11%). A unique finding of this study was that only one day of strength training combined with another day of endurance training performed on a cycle ergometer resulted in enhanced muscle mass in the elderly after 16 weeks.

In another study, Karavirta et al. [39], have shown increase in the cross-sectional area of type II muscle fibers of the vastus lateralis only in the strength training group (~16%), whereas no changes were observed in the concurrent training group. However, this difference did not result in different strength gains. In other studies, utilizing a weekly training frequency ranging from two to three times with intensities of 40–80% of 1 RM (progressive load during training periodization) with multiple sets produced marked increases in muscle mass (9–16%), with no differences between the strength and concurrent training interventions [34, 35, 37–39]. Moreover, although the intra-session exercise sequence influenced strength adaptations, it is important to note that the sequence of strength and endurance exercise had no influence on muscle mass gains [28, 31]. Thus, independent of the intra-session exercise sequence, the performance of endurance training does not impair the hypertrophy induced by strength training when both types of training are

performed simultaneously. As observed with muscle strength, Ferrari et al. [30] have shown similar muscle hypertrophy comparing a weekly frequency of three times a week with only twice weekly training sessions (approximately 5% in the quadriceps muscles) in previously concurrently trained elderly subjects. Regarding the use of maximal or submaximal sets (i.e., repetitions to failure or not to failure) during concurrent training in elderly, da Silva et al. [33] compared three types of concurrent training interventions in healthy elderly men: one with the strength training performed with repetitions to concentric failure (RFG); another with strength training performed with 50% of the repetitions to concentric failure (NFG); and a third, with strength training performed with 50% of the repetitions to concentric failure, but equalizing the total volume by adding more sets. We demonstrated that the groups that performed a higher training volume showed greater muscle hypertrophy, although repetitions to failure did not provide additional muscle size gains compared to repetitions not to failure using the same total load [33].

Do Changes in Anabolic and Catabolic Hormone Concentrations Explain the Interference Effect in the Elderly?

The results regarding the effects of concurrent training on muscle mass enhancements in the elderly are in agreement with the studies conducted in young populations, as no difference was observed between the strength training and concurrent training groups when imaging techniques were used to evaluate these effects [44, 45]. However, studies that used muscle biopsies demonstrated the interference phenomenon in the cross-sectional area of type I fibers in young men who performed concurrent training with a high intensity and volume of both strength and endurance training [13, 46]. The interference occurred concomitantly with increases in cortisol levels, suggesting that these subjects may have been in an exacerbated catabolic state [13, 47]. In the elderly, no evidence of an overtraining state has been observed. In the study by Karavirta et al. [39], a moderate volume of training resulted in an increase in the type II fiber area only in the strength training group, while no changes were observed in the concurrent training group. Notwithstanding, no changes in the basal anabolic or catabolic hormone levels were evidenced in this study [39]. In addition, in a previous study of our laboratory, Cadore et al. [25] have shown that the interference effect on neuromuscular adaptations occurred with no changes in the levels of free and total testosterone or cortisol, suggesting that the interference phenomenon occurred with no evidence of an overtraining state in the elderly.

Effects of Concurrent Training on Maximal and Submaximal Neuromuscular Activity

Early adaptations to strength training in the elderly include increases in the maximal neuromuscular activity (i.e., EMG signal amplitude) [48–52]. These changes may suggest the occurrence of neural adaptations, such as increases in the maximal

motor unit recruitment [15], maximal motor unit firing rate [16], spinal motorneuronal excitability, and efferent motor drive [17]. Although neural adaptation impairments are suggested as a mechanism to explain the interference effect, few studies have investigated the neural adaptations to concurrent training. In elderly subjects, the interference effect may be related to impaired neural adaptations to strength training, since the lower strength gains induced by concurrent training when compared with strength training alone may occur in parallel with lower maximal neuromuscular activity adaptations induced by concurrent training [25].

When investigating the mechanisms underlying the strength adaptations to strength and concurrent training in the elderly, we observed a significant increase in the maximal EMG amplitude of the *rectus femoris* and *vastus lateralis* only in the strength training group (~30%), and these modifications were significantly greater than those observed in the concurrent training group (~1.5%, not statistically significant) [25]. In addition, we also observed greater isometric neuromuscular economy (reduced submaximal EMG to the same absolute load after training) in the *rectus femoris* and *vastus lateralis* only in the strength training group. It is important to highlight that the greater neuromuscular activity changes observed in strength training than concurrent training occurred in parallel with greater strength gains in strength training [25], which suggested that the interference effect occurred at least in part due to impairments in neural changes.

In another study, Cadore et al. [27] found significantly greater changes in the force per unit of active muscle mass (i.e., muscle quality or specific tension) in elderly individuals who performed strength training prior to an endurance exercise sequence when compared with the inverse order (27 vs. 15%, $P < 0.01$). The force per unit of active muscle mass provides an estimation of the contribution of neuromuscular factors associated with changes in strength development, as enhanced strength with the same muscle mass suggests neural adaptations to training [53–55]. In addition, Cadore et al. [28] showed greater changes in the neuromuscular economy of the *rectus femoris* in elderly individuals who performed strength training prior to an endurance exercise sequence when compared with the inverse order. Taken together, these results suggest that the interference effect in the elderly might be explained at least in part by impairments in the neural adaptations to strength training. From a practical point of view, although the concurrent training performance may impair the neuromuscular adaptations, the performance of strength prior to endurance intra-session exercise sequence seems to minimize this negative effect and should be considered to optimize the benefits of concurrent training in the neuromuscular function in elderly.

Effects of Concurrent Training on Cardiorespiratory Adaptations

Along with the decrease in the maximal cardiac output [56, 57], several authors have demonstrated that the cardiorespiratory fitness declines are also associated with strength and power decreases related with aging [1, 2, 27, 58]. In line with this, some studies have shown that the combination of strength and endurance training is a

better strategy to improve the cardiorespiratory fitness of the elderly when compared with endurance and especially strength training alone. In addition, the performance of strength training simultaneously with endurance training does not impair the aerobic adaptations produced by endurance training alone [21, 22, 34–37, 39, 58, 59].

Studies that have investigated cardiovascular adaptations to CT have demonstrated increases ranging from 10 to 18% in the maximum oxygen uptake and maximal cycle ergometer workload in elderly individuals who underwent training periods ranging from 12 to 21 weeks, and a weekly frequency ranging from two to three training sessions [21, 22, 24, 37–39, 58, 59]. Similar to the results observed in strength performance and hypertrophy mentioned above, Izquierdo et al. [22] observed similar aerobic power gains in elderly men who underwent one session per week of strength training and one session per week of cycle endurance training in the CT group (28%) and those who underwent ET twice per week (23%) after 16 weeks of training.

Interestingly, in the study by Cadore et al. [27], similar enhancements were observed in the peak oxygen uptake, maximal workload at cycle ergometer, and the workload at the second ventilatory threshold among groups that performed strength training prior to an endurance exercise sequence and the opposite exercise order. However, a greater improvement was found in the workload at the first ventilatory threshold in the group that performed strength training prior to endurance exercise in each session. It is possible that this difference was observed as a consequence of the greater increases in muscle strength achieved by performing strength training prior to endurance training, as strength gains have been associated with maximal and submaximal endurance gains [2] and dynamic neuromuscular economy in the elderly [24]. In another study, we have shown that, independent of the exercise order (i.e., SE or ES), 12 weeks of concurrent training significantly reduced the submaximal oxygen uptake at the highest submaximal intensity (i.e., 100 W) in elderly men [60]. Based on this study, we also concluded that the prevalence of non-responders in the maximal aerobic power was lower in the elderly group who performed endurance training regularly before strength training [60]. Regarding different weekly training frequencies, the aforementioned study of Ferrari et al. [30] showed similar $\text{VO}_{2\text{peak}}$ increases after 10 weeks of concurrent training performed two or three times a week in well-trained elderly men. However, greater maximal power at cycle ergometer (W_{max}) gains were observed in the group who trained three times per week, suggesting that in trained elderly, a greater weekly training frequency may be necessary to promote overall endurance gains.

Effects of Concurrent Training on Functional Capacity

Few studies have compared the effects of strength, endurance, and concurrent training on functional capacity in the elderly. In an investigation by Wood et al. [21], no differences were observed between the strength-, endurance-, and concurrent training groups in functional performance gains assessed by the sit and reach test, agility/dynamic balance assessed by repeatedly standing from a chair, walking around cones and returning to the chair (i.e., TUG test), and coordination tests. Holviala

et al. [59] showed increases in the treadmill load carrying walking test performance (10.1 kg in each hand) only in the concurrent training group (4.5%), whereas no changes were observed in the strength training and endurance training groups. In another study, Wilhelm et al. [31] have shown that concurrent training led to improved sit-to-stand ability (i.e., number of repetitions in 30 s), as well as timed-up-and-go test performance in healthy elderly men, independent of the exercise sequence. Altogether, these results suggested that the combination of strength and endurance capacities improves functional capacity in the elderly population. However, it should be also mentioned that in the oldest aged individuals, especially those individuals with frailty syndrome or at greater functional declines, the exercise intervention approach must combine balance and gait exercises along with strength and endurance stimuli, in order to improve the functional capacity and reduce the risk of falls [29, 61, 62]. In addition, in view of improving the concurrent strength and endurance training prescription, strength training should be prescribed combining slow and explosive mode contractions in order to optimize the functional capacity enhancements. This recommendation is because maximal muscle power output and rate of force development are more associated with functional capacity than maximal strength per se [12, 63, 64].

Summary

Strength training is an effective intervention to improve muscle strength, power output, and muscle mass in the elderly. Endurance training induces improvements in VO_{2max} and submaximal endurance capacity in these populations. Therefore, a combination of strength and endurance training (i.e., concurrent training) seems to be the most effective way to improve both neuromuscular and cardiorespiratory functions. Based on recent evidence, concurrent training performed at a moderate weekly frequency (i.e., two times per week) may promote marked gains in muscle hypertrophy, strength and power gains in elderly subjects. The strength training should be performed at moderate to high intensity (i.e., 60–80% of 1RM), and moderate volume (i.e., 2–3 sets per exercise). Each session should target the main muscle groups of the body. Also, endurance training should be performed at moderate to high intensity (i.e., 60–85% of VO_{2max}), and last for 25–40 min. For concurrent training protocols in which both strength and endurance training are performed on the same day, the strength and endurance gains may be optimized with strength training being performed prior to endurance exercise. Moreover, two weekly sessions of both strength and endurance training may be an optimal weekly frequency to promote additional muscle mass and strength gains, as well as cardiorespiratory fitness in previously concurrent trained elderly. Furthermore, performing repetitions until concentric failure does not provide further neuromuscular performance gains and muscle hypertrophy compared to a lower number of repetition in this population. In terms of improving the functional capacity of the elderly, the concurrent strength and endurance training prescription should include high-velocity strength training, designed to improve muscle power output, as muscle power has been associated with the functional capacity.

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